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Chemical Bonding: Hybridisation and Spatial Geometry:

Source questions

Q1 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Electron-Domain Counting and Molecular Shape Geometry

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Steric Number and Hybridisation in Molecules with Lone Pairs

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

BF₃ and NH₃ each have three sigma bonds to the central atom. BF₃ has no central lone pair; NH₃ has one. Explain why three

Representation

sigma bonds alone cannot determine the same central hybridisation and molecular shape for both.

Common wrong route · open if you want a warning

+

Supplied three-bond comparison



BF_3 Lewis inventory

NH_3 Lev

Connectivity layout only; page angles are not measured ge
Supplied situation

Need a hint? · guided support

+

Need the concept again?

Electron-Domain Counting and Molecular Shape Geometry

Q2 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Multiple Bonds, Orthogonal Pi Systems, and Terminal Atom Hybridisation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

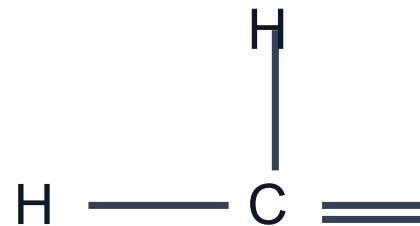
For $\text{H}_2\text{C}=\text{O}$, assign the carbon hybridisation, count the sigma and pi components of the $\text{C}=\text{O}$ bond, and identify the carbon orbital left available for the pi component. Keep electron-domain counting separate from counting shared electron pairs.

Representation

**Common wrong route · open if you
want a warning**

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H₂C=O connectivity



H₂C=O connectivity

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

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Need the concept again?

Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

Q3 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Multiple Bonds, Orthogonal Pi Systems, and Terminal Atom Hybridisation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Ethene is $\text{H}_2\text{C}=\text{CH}_2$. Translate this Lewis representation into a qualitative orbital-overlap diagram showing the C–C sigma framework, the two unhybridised carbon p orbitals and their side-on overlap. Label which directions are in the molecular plane and which are perpendicular to it.

Representation

**Common wrong route · open if you
want a warning**

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Ethene Lewis connectivity



Ethene connectivity

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

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Need the concept again?

Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

Q4 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Multiple Bonds, Orthogonal Pi Systems, and Terminal Atom Hybridisation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For CO₂, connect the central-carbon Lewis domain count to its hybrid-orbital model, the number of remaining carbon p orbitals and the sigma/pi composition of its two C=O bonds. Explain how these pieces fit one consistent linear arrangement.

Representation

**Common wrong route · open if you
want a warning**

+

CO₂ Lewis connectivity



CO₂ connectivity

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

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Need the concept again?

Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

Q5 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Electron-Domain Counting and Molecular Shape Geometry

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Steric Number and Hybridisation in Molecules with Lone Pairs

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

Consider the claim: 'Any central atom with three sigma bonds must be sp^2 .' Given BF_3 : three B–F sigma bonds, zero boron lone pairs, planar; NH_3 : three N–H sigma bonds, one nitrogen lone pair, pyramidal. Decide whether these observations support the claim, give the decisive

Representation

warrant, and state a more defensible introductory rule. Explain what the shape observations alone do not prove about orbitals.

Common wrong route · open if you want a warning

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Supplied BF₃ and NH₃ counterexample candidate



BF₃ Lewis inventory

NH₃ Lev

Connectivity layout only; page angles are not measured ge
Supplied situation

Need a hint? · guided support

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Need the concept again?

Electron-Domain Counting and Molecular Shape Geometry

Q6 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Electron-Domain Counting and Molecular Shape Geometry

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Steric Number and Hybridisation in Molecules with Lone Pairs

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

In CH₃F, carbon has three C–H single bonds and one C–F single bond. Assign its introductory local hybridisation and explain why the number of electron-domain directions is unchanged from CH₄ even though the attached atoms are

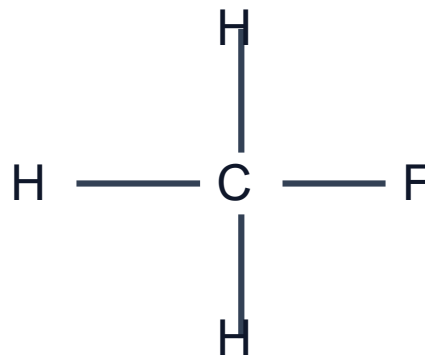
Representation

different. Do not assume all bond properties are identical.

Common wrong route · open if you want a warning

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CH₃F and CH₄ connectivity



CH₃F connectivity



CH₄ con

Connectivity layout only; page angles are not measured ge
Supplied situation

Need a hint? · guided support

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Need the concept again?

Electron-Domain Counting and Molecular Shape Geometry

Q7 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Multiple Bonds, Orthogonal Pi Systems, and Terminal Atom Hybridisation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

HCN is represented as $\text{H}-\text{C}\equiv\text{N}$ with a lone pair on nitrogen. Assign the introductory hybridisation at carbon and nitrogen, and count the sigma and pi components of the $\text{C}\equiv\text{N}$ bond. Explain how the local domain count is compatible with those components.

Representation

**Common wrong route · open if you
want a warning**

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Supplied HCN Lewis connectivity



HCN connectivity; : is the 1

Connectivity layout only; page angles are not meas

Supplied situation

Need a hint? · guided support

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Need the concept again?

Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

Q8 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Electron-Domain Counting and Molecular Shape Geometry

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Steric Number and Hybridisation in Molecules with Lone Pairs

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

For methylamine, $\text{CH}_3\text{-NH}_2$, carbon has four single bonds and nitrogen has three single bonds plus a lone pair. Compare the introductory local hybridisation and arrangement of bonded atoms at carbon and nitrogen. Explain why the same

Representation

hybridisation label does not require identical molecular shapes.

Common wrong route · open if you want a warning

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Supplied methylamine connectivity



Nitrogen lone pair is n

Connectivity layout only; page angles are not meas

Supplied situation

Need a hint? · guided support

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Need the concept again?

Electron-Domain Counting and Molecular Shape Geometry

Q9 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Multiple Bonds, Orthogonal Pi Systems, and Terminal Atom Hybridisation

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

A carbon centre has three sigma-bond directions, no lone pair, and participates in one pi bond. Translate this description into a qualitative local orbital arrangement and check that the orbitals used in the sigma and pi parts have not been counted twice.

Representation

**Common wrong route · open if you
want a warning**

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Supplied local carbon jobs

Three sigma directions; no lone pair

One pi participation

Available valence basis: one 2s and three

Connectivity layout only; page angles are not measured

Supplied situation

Need a hint? · guided support

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Need the concept again?

Multiple Bonds, Unhybridised Orbitals, and Spatial Coordination

Q10 Owner-supplied question, verbatim · coordinator/AI-drafted benchmark

Subject: Chemistry

Topic: Chemical Bonding and Molecular Structure

Concept: Electron-Domain Counting and Molecular Shape Geometry

Concept difficulty: Medium concept

Question difficulty: Connected reasoning

Family: Steric Number and Hybridisation in Molecules with Lone Pairs

Question type: Constructed response

Source: Owner-supplied question

Provenance: Owner-supplied question

ATTEMPT FIRST

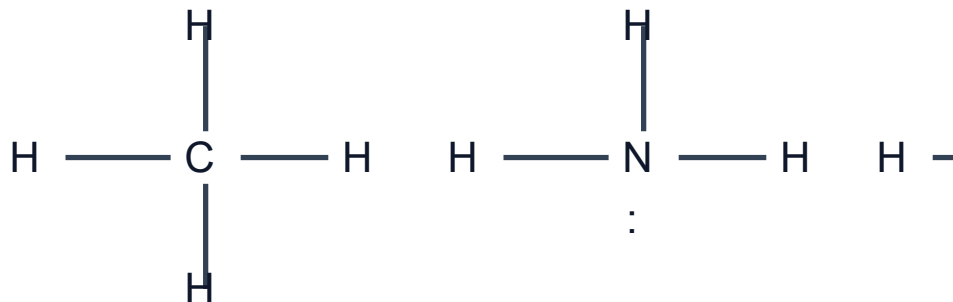
A student says, 'If the electron-domain geometry is tetrahedral, the atoms must form a tetrahedral molecular shape.'
Evaluate the statement using CH₄, NH₃ and H₂O, with their respective central lone-pair counts of zero, one and two.

Representation

Common wrong route • open if you want a warning

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Supplied four-domain comparisons



CH₄ connectivity

NH₃ Lewis inventory

H₂O I

Connectivity layout only; page angles are not measured geometry.

Supplied situation

Need a hint? • guided support

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Need the concept again?

Electron-Domain Counting and Molecular Shape Geometry

